1.11 Quadratic Modelling

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Modelling

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The new A Level has a particular emphasis of the application of theory to real-life situations. A mathematical model is the maths used to model such a situation, possibly with some simplifying assumptions.

Example (from textbook): A spear is thrown over level ground from the top of a tower. The height, in metres, of the spear above the ground after t seconds is modelled by the function: $h(t) = 12.25 + 14.7t - 4.9t^2, \quad t \geq 0$

a) Interpret the meaning of the constant term 12.25 in the model.
b) After how many seconds does the spear hit the ground?
c) Write $h(t)$ in the form $A - B(t - C)^2$, where A, B and C are constants to be found.
d) Using your answer to part c or otherwise, find the maximum height of the spear above the ground, and the time at which this maximum height is reached?

a

b

c

d

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2 A car manufacturer uses a model to predict the fuel consumption, y miles per gallon (mpg), for a specific model of car travelling at a speed of x mph. $y = -0.01x^2 + 0.975x + 10, \ x > 0$.

a Use the model to find two speeds at which the car has a fuel consumption of 32.5 mpg.
b Rewrite y in the form $A - B(x - C)^2$, where A, B, C are constants to be found.
c Using your answer to part b, find the speed at which the car has the greatest fuel efficiency.
d Use the model to calculate the fuel consumption of a car travelling at 120 mph.
Comment on the validity of using this model for very high speeds.

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3 A fertiliser company uses a model to determine how the amount of fertiliser used, f kilograms per hectare, affects the grain yield y, measured in tonnes per hectare. $y = 4 + 0.3f - 0.0006f^2$.

a According to the model, how much grain would each hectare yield without any fertiliser? (1 mark)

b One farmer currently uses 20 kilograms of fertiliser per hectare. How much more fertiliser would he need to use to increase his grain yield by 1 tonne per hectare? (4 marks)

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4 A football stadium has 25000 seats. The football club know from past experience that they will sell only 10000 tickets if each ticket costs $\pounds 30$. They also expect to sell 1000 more tickets every time the price goes down by $\pounds 1$. **a** The number of tickets sold t can be modelled by the linear equation $t = M - 1000p$, where $\pounds p$ is the price of each ticket and M is a constant. Find the value of M. (1 mark)

The total revenue, $\pounds r$, can be calculated by multiplying the number of tickets sold by the price of each ticket. This can be written as $r = p(M - 1000p)$.

b Rearrange r into the form $A - B(p - C)^2$, where A, B, C are constants to be found. (3 marks)

c Using your answer to part b or otherwise, work out how much the football club should charge for each ticket if they want to make the maximum amount of money. (2 marks)